Exam 1 for MATH 421

March 2, 2015

NAME (*please print*): _____

SIGNATURE: _____

Do all problems, in any order.

Show all your work. Full credit may not be given for an answer alone.

You may use <u>no</u> course materials.

Problem Number	Possible Points	Points Earned:
1	20	
2	20	
3	10	
4	10	
5	10	
6	10	
7	20	
Total Points Earned:		

(20) 1. a) Find the Laplace transform of $2e^{2t}\delta(t-2) - 5e^{6t}\cos(5t)$.

$$2e^{-2(s-2)} - 5\frac{s-6}{(s-6)^2+25}$$

b) Find the inverse Laplace transform of $\frac{10s}{(s^2+25)^2}\,e^{-5s}$.

$$(t-5)\sin 5(t-5)\,\mathcal{U}(t-5)$$

(20) 2. Find the inverse Laplace transform of $\frac{1}{(s^2+1)(s+1)}$ using the following methods:

a) Partial fraction.

$$\frac{1}{2}\sin t - \frac{1}{2}\cos t + \frac{1}{2}e^{-t}$$

b) The convolution theorem.

(10) 3. Solve the following initial value problem:

$$\begin{cases} y'' + y = \mathcal{U}(t-4), \\ y(0) = 2, \quad y'(0) = 0. \end{cases}$$

$$y = 2\cos t + \left(-\cos(t-4) + 1\right)\mathcal{U}(t-4)$$

(10) 4. Solve the following initial value problem using the method of Laplace transform:

$$\begin{cases} y_1' + y_2 = 0, \\ y_2' - y_1 = 0 \\ y_1(0) = 1, \quad y_2(0) = 0. \end{cases}$$

$y_1 = \cos t$

$y_2 = \sin t$

(10) 5. Calculate the determinant

$$\det \begin{pmatrix} 2 & 2 & 2 & 4 \\ 4 & 2 & 2 & 5 \\ 1 & 1 & 3 & 0 \\ -1 & -1 & 1 & -5 \end{pmatrix}.$$

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(10) 6. Let $A = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 1 \\ 3 & 1 & -1 \end{pmatrix}$. Determine whether A is invertible. If A is invertible, find A^{-1}.

Invertible.

$$A^{-1} = \begin{pmatrix} -3 & 4 & 1 \\ 4 & -5 & -1 \\ -5 & 7 & 1 \end{pmatrix}$$

(20) 7. Let $A = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 3 & -1 \\ 3 & -2 & 2 \end{pmatrix}$.

a) Calculate the rank of A.

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b) Find all the eigenvalues and eigenvectors of A.

$\lambda = 1, 2, 4$

(1) $\lambda = 1$, $X_2 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$. $X_2 \neq 0$

(2) $\lambda = 2$, $X_3 \begin{pmatrix} -2 \\ -3 \\ 1 \end{pmatrix}$, $X_3 \neq 0$

(3) $\lambda = 4$, $X_3 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$, $X_3 \neq 0$

Exam 2 for MATH 421

April 13, 2015

NAME (*please print*): _____

Do all problems, in any order.

Show all your work. Full credit may not be given for an answer alone.

You may use no course materials.

Problem Number	Possible Points	Points Earned:
1	10	
2	20	
3	10	
4	20	
5	15	
6	5	
7	20	
Total Points Earned:		

(10) 1. Let $A = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$. Find a diagonal matrix D and an orthogonal matrix P such that $P^T AP = D$.

$$P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 6 \end{pmatrix}$$

(20) 2. Let

$$f(x) = \begin{cases} 2 & -\pi \le x \le -\pi/2 \\ 0 & -\pi/2 < x < \pi/2 \\ -3 & \pi/2 \le x \le \pi \end{cases}$$

a) Find the full Fourier series of $f(x)$.

$$a_0 = -\frac{1}{2}$$

$$a_n = \frac{1}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = \frac{5}{n\pi}\left((-1)^n - \cos\frac{n\pi}{2}\right)$$

The full Fourier series of f(x):

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right)$$

b) Is the Fourier series convergent when $x \neq -\pi, -\pi/2, \pi/2, \pi$? If it is convergent when $x \neq -\pi, -\pi/2, \pi/2, \pi$, what is its limit at this point?

Convergent to $f(x)$.

c) Is the Fourier series convergent when $x = -\pi$, $x = -\pi/2$ or $x = \pi/2$ or $x = \pi$? If it is convergent at $x = -\pi$ or $x = -\pi/2$ or $x = \pi/2$ or $x = \pi$, what is the limit at this point?

Convergent to

$\frac{2-3}{2} = -\frac{1}{2}$ at $x = -\pi, \pi$;

$\frac{2+0}{2} = 1$ at $x = -\frac{\pi}{2}$

$\frac{0-3}{2} = -\frac{3}{2}$ at $x = \frac{\pi}{2}$

(10) 3. Let

$$f(x) = \begin{cases} x & 0 < x < \pi/2 \\ 0 & \pi/2 \le x \le \pi. \end{cases}$$

Find the Fourier series of $f(x)$ using the even extension of $f(x)$.

The Fourier series is $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$

where

$$a_0 = \frac{\pi}{4}$$

$$a_n = \frac{1}{n} \sin \frac{n\pi}{2} + \frac{2}{n^2 \pi} \cos \frac{n\pi}{2} - \frac{2}{n^2 \pi}$$

(20) 4. Solve the eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 \le x < 2\pi, \\ X'(0) = 0, X'(2\pi) = 0. \end{cases}$$

That is, find all the eigenvalues λ and the corresponding eigenfunctions (nonzero solutions) X corresponding to λ.

$$\lambda_n = \left(\frac{n}{2}\right)^2, \quad n = 0, 1, 2, 3, \ldots$$

$$X_n = A_n \cos \frac{nx}{2}$$

(15) 5. Find all the solutions of the equation $y^3 \frac{\partial u}{\partial x} = x^5 \frac{\partial u}{\partial y}$ of the form $u(x, y) = X(x)Y(y)$ (that is, all separable or product solutions).

$$u = C e^{-\lambda\left(\frac{1}{4}x^{-4} + \frac{1}{2}y^{-2}\right)}$$

(5) 6. What is the type of the equation

$$2\frac{\partial^2 u}{\partial x^2} - 7\frac{\partial^2 u}{\partial x \partial y} - 3\frac{\partial^2 u}{\partial y^2} = 0.$$

Hyperbolic

(20) 7. Solve the problem

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = \dfrac{\partial^2 u}{\partial x^2}, & 0 < x < \pi, 0 < t < \infty \\ u(0,t) = u(\pi,t) = 0, & 0 < t < \infty \\ u(x,0) = 10\sin(21x),\ u_t(x,0) = -3\sin(3x) & 0 < x < \pi. \end{cases}$$

(Please give the solution explicitly.)

$$u = XT \Rightarrow XT'' = X''T \Rightarrow \frac{X''}{X} = \frac{T''}{T} = -\lambda$$

$$X'' + \lambda X = 0$$

$$T'' + \lambda T = 0$$

$$u(0,t) = u(\pi,t) = 0 \Rightarrow X(0) = X(\pi) = 0$$

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(\pi) = 0 \end{cases} \Rightarrow \lambda = \lambda_n = n^2 \quad X = X_n = A_n \sin nx$$

for $n = 1, 2, 3, \cdots$

$$T_n = B_n \cos nt + C_n \sin nt$$

$$u_n = X_n T_n = a_n \cos nt \sin nx + b_n \sin nt \sin nx$$

where $a_n = A_n B_n$, $b_n = A_n C_n$.

$$u = \sum_{n=1}^{\infty} (a_n \cos nt \sin nx + b_n \sin nt \sin nx)$$

$$u_t = \sum_{n=1}^{\infty} (-a_n n \sin nt \sin nx + b_n n \cos nt \sin nx)$$

$$10 \sin 21x = u(x,0) = \sum_{n=1}^{\infty} a_n \sin nx \Rightarrow a_n = 0,\ n \neq 21$$

$$a_{21} = 10$$

$$-3 \sin 3x = u_t(x,0) = \sum_{n=1}^{\infty} b_n n \sin nx \Rightarrow b_n n = 0,\ n \neq 3$$

or $b_n = 0,\ n \neq 3$

$b_3 3 = -3$

$\Rightarrow b_3 = -1$

$$u(x,t) = 10 \cos 21t \sin 21x - \sin 3t \sin 3x$$